



## QUESTIONS

Q1

How many edges must a Minimum Spanning Tree of a graph with  $SV5$  vertices have?  
2 marks



Q2

In a DAG, which node will always have an in-degree of zero in a topological ordering?  
2 marks



Q3

Differentiate between directed and undirected graphs.  
2 marks



Q4

Define a connected graph.  
2 marks



Q5

What is a topological sort?  
2 marks



5/5 answered

Q4 Define a connected graph.

2 marks

Define a connected graph.

Your Answer

a connected graph is a graph in which there exists a path between every pair of vertices. this means that starting from any vertex it is possible to reach every other vertex in the graph by traversing along the edges.

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2 marks



5/5 answered

Q2 In a DAG, which node will always have an in-degree of zero in a topological ordering?

2 marks

In a DAG, which node will always have an in-degree of zero in a topological ordering?

Your Answer

the source node will always have an in degree of zero in a topological ordering of a dag. In a topological sort this node must appear first since it has no dependencies on any other node before it can be processed.

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5/5 answered

Q5 What is a topological sort?

2 marks

What is a topological sort?

Your Answer

topological sort is a linear ordering of the vertices of a directed acyclic Graph such that for every directed edge  $u \rightarrow v$  vertex  $u$  comes before vertex  $v$  in the ordering. It is used to schedule tasks based on their dependencies and it only exists for graphs that have no cycles.

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5/5 answered

Q1 How many edges must a Minimum Spanning Tree of a graph with  $V$  vertices have?

2 marks

How many edges must a Minimum Spanning Tree of a graph with  $V$  vertices have?

Your Answer

MINIMUM SPANNING TREE:

a minimum spanning tree of a graph with  $v$  vertices must have exactly  $v-1$  edges. this is because a spanning tree connects all  $v$  vertices without forming any cycles and a tree with  $V$  vertices always has exactly  $v-1$  edges.

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5/5 answered

Q3 Differentiate between directed and undirected graphs.

2 marks

Differentiate between directed and undirected graphs.

Your Answer

In a directed graph each edge has a specific direction meaning it goes from one vertex to another and traversal is only possible in that direction

In an undirected graph edges have no direction, meaning the connection between two vertices works both ways

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